

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459369.9295 +/- 0.0066 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.045
Transit depth (Rp/Rs)^2	1.94 +/- 1.26 [%]
Orbital Inclination (inc)	83.49 +/- 2.74
Airmass coefficient 1 (a1)	1.066 +/- 0.028
Airmass coefficient 2 (a2)	-0.046 +/- 0.034
Scatter in the residuals of the lightcurve fit is	2.68 %
Best Comparison Star	#2 - [430, 233]
Optimal Aperture	4.79
Optimal Annulus	11.18
Transit Duration (day)	0.046 +/- 0.035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.139 ± 0.045 vs 0.1326 (deviation 0.09σ)
Duration fit vs published	0.046 d vs 0.0737 d (deviation 0.52σ)
Mid-time O–C	16.4 min
Depth SNR	2.0
Residual scatter	2.68 %

Light curve

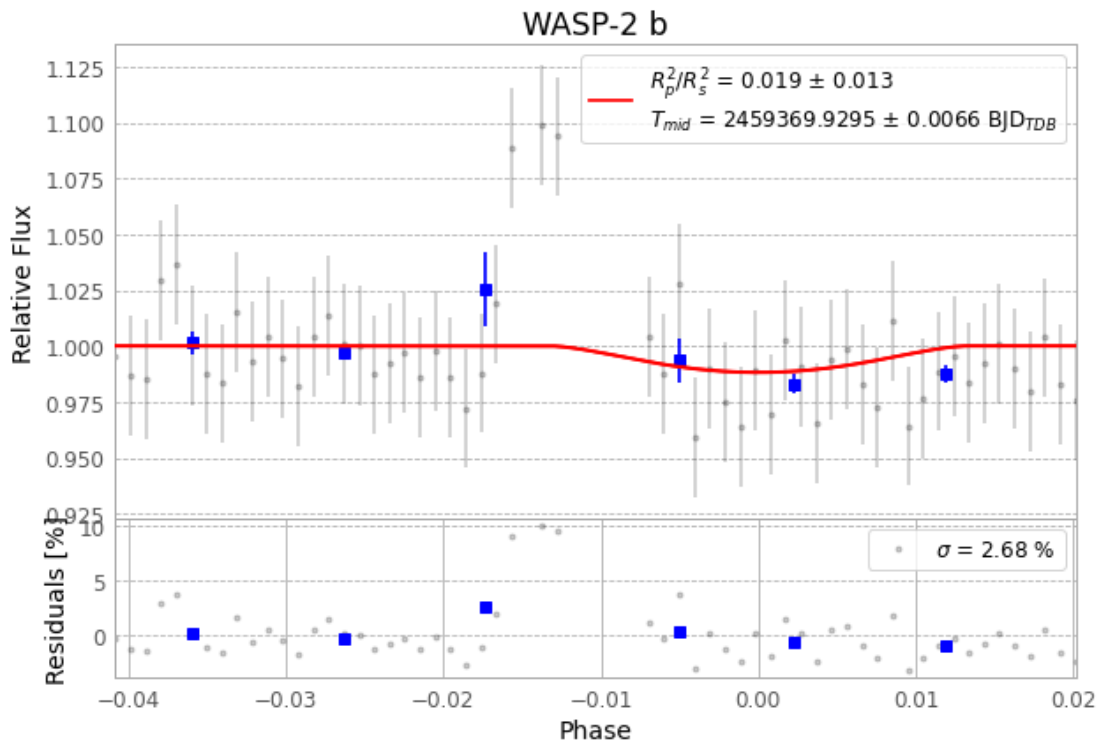


Figure 1 — FinalLightCurve_WASP-2 b_2021-06-04.png (SHA-256 4e366fc2ef5803ce...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [388, 116], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2bea967d9933a7fe...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), WASP-2210604080612.FITS → WASP-2210604111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-210604.log (eb3f8964bdeb...), FinalLightCurve_WASP-2 b_2021-06-04.png (4e366fc2ef58...), FinalLightCurve_WASP-2 b_2021-06-04.csv (623b77ecb79a...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 18:37Z.